Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Contents

Saturday 11 April 2026

1.1 Daily Limit

Limit of Rational Polynomial Composition Ratio

Let $P(x) = x^3 + 4x^4 + 3x^5 + 4x^6$. Find the value of:

$$\lim_{n \to 1} \frac{P(n^6 - 1)}{P(n^7 - 1)}$$

Solution: Let $u = n - 1$. As $n \to 1$, we have $u \to 0$. We begin by expanding the arguments $n^6 - 1$ and $n^7 - 1$ in terms of u:

$$n^6 - 1 = (1 + u)^6 - 1 = 6u + O(u^2)$$

$$n^7 - 1 = (1 + u)^7 - 1 = 7u + O(u^2)$$

Next, analyze the behavior of $P(x)$ near $x = 0$. Since $P(x) = x^3(1 + 4x + 3x^2 + 4x^3)$, the leading term of $P(x)$ as $x \to 0$ is x^3:

$$P(x) = x^3 + O(x^4) \quad \text{as } x \to 0$$

Substituting $x = 6u + O(u^2)$ and $x = 7u + O(u^2)$ into $P(x)$:

$$P(n^6 - 1) = (6u + O(u^2))^3 + O(u^4) = 216u^3 + O(u^4)$$

$$P(n^7 - 1) = (7u + O(u^2))^3 + O(u^4) = 343u^3 + O(u^4)$$

Taking the limit as $u \to 0$:

$$L = \lim_{u \to 0} \frac{216u^3 + O(u^4)}{343u^3 + O(u^4)} = \frac{216}{343} = \left(\frac{6}{7}\right)^3$$

Sunday 12 April 2026

2.1 Daily Limit

Exponential Power Limit of Harmonic Ratio

Evaluate the limit:

$$\lim_{x\to\infty}\left(\frac{\sum_{k=1}^{x}\frac{1}{k}}{\ln x}\right)^{\ln x}$$

Solution: Let L denote the limit. The summation in the numerator is the x-th harmonic number H_x, which has the asymptotic expansion as $x \to \infty$:

$$H_x = \ln x + \gamma + O\left(\frac{1}{x}\right)$$

where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant. Substituting this into the base of our exponential expression:

$$\frac{H_x}{\ln x} = \frac{\ln x + \gamma + O(1/x)}{\ln x} = 1 + \frac{\gamma}{\ln x} + O\left(\frac{1}{x\ln x}\right)$$

Taking the natural logarithm of the expression $y(x) = \left(\frac{H_x}{\ln x}\right)^{\ln x}$:

$$\ln y(x) = \ln x \cdot \ln\left(1 + \frac{\gamma}{\ln x} + O\left(\frac{1}{x\ln x}\right)\right)$$

Using the Taylor expansion $\ln(1+t) = t - \frac{t^2}{2} + O(t^3)$ as $t \to 0$, with $t = \frac{\gamma}{\ln x}$:

$$\ln y(x) = \ln x\left(\frac{\gamma}{\ln x} + O\left(\frac{1}{(\ln x)^2}\right)\right) = \gamma + O\left(\frac{1}{\ln x}\right)$$

Taking the limit as $x \to \infty$:

$$\lim_{x\to\infty} \ln y(x) = \gamma$$

Exponentiating both sides yields the final limit:

$$L = e^{\gamma}$$

Monday 13 April 2026

3.1 Daily Limit

Asymptotic Limit Involving Gamma Function

Evaluate the limit:

$$\lim_{x\to\infty} \frac{x^{x-\frac{1}{2}}}{\Gamma(x)e^x}$$

Solution: We recall Stirling's approximation formula for the Gamma function as $x \to \infty$:

$$\Gamma(x) \sim \sqrt{\frac{2\pi}{x}} \left(\frac{x}{e}\right)^x = \sqrt{2\pi}\, x^{x-\frac{1}{2}} e^{-x}$$

Substituting this expansion directly into the denominator of our quotient:

$$\Gamma(x)e^x \sim \sqrt{2\pi}\, x^{x-\frac{1}{2}} e^{-x} \cdot e^x = \sqrt{2\pi}\, x^{x-\frac{1}{2}}$$

Now substitute this result into the limit expression:

$$L = \lim_{x\to\infty} \frac{x^{x-\frac{1}{2}}}{\sqrt{2\pi}\, x^{x-\frac{1}{2}}} = \frac{1}{\sqrt{2\pi}} = \frac{\sqrt{2\pi}}{2\pi}$$

Tuesday 14 April 2026

4.1 Daily Limit

Limit of Scaled Beta Function Ratio

Evaluate the limit:

$$\lim_{n\to\infty} \frac{\mathrm{B}\left(\frac{1}{n}, 1+\frac{1}{n}\right)}{n}$$

Solution: We express the Beta function in terms of the Gamma function:

$$\mathrm{B}(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

Setting $a = \frac{1}{n}$ and $b = 1 + \frac{1}{n}$:

$$\mathrm{B}\left(\frac{1}{n}, 1+\frac{1}{n}\right) = \frac{\Gamma\left(\frac{1}{n}\right)\Gamma\left(1+\frac{1}{n}\right)}{\Gamma\left(1+\frac{2}{n}\right)}$$

Using the functional equation $\Gamma(z+1) = z\Gamma(z)$ for $z = \frac{1}{n}$, we have $\Gamma\left(\frac{1}{n}\right) = n\Gamma\left(1+\frac{1}{n}\right)$. Substituting this gives:

$$\mathrm{B}\left(\frac{1}{n}, 1+\frac{1}{n}\right) = \frac{n\left[\Gamma\left(1+\frac{1}{n}\right)\right]^2}{\Gamma\left(1+\frac{2}{n}\right)}$$

Dividing by n cancels the factor in front:

$$\frac{\mathrm{B}\left(\frac{1}{n}, 1+\frac{1}{n}\right)}{n} = \frac{\left[\Gamma\left(1+\frac{1}{n}\right)\right]^2}{\Gamma\left(1+\frac{2}{n}\right)}$$

By continuity of the Gamma function and since $\Gamma(1) = 1$:

$$\lim_{n\to\infty} \Gamma\left(1+\frac{1}{n}\right) = \Gamma(1) = 1$$

$$\lim_{n\to\infty} \Gamma\left(1+\frac{2}{n}\right) = \Gamma(1) = 1$$

Therefore, the limit is:

$$L = \frac{1^2}{1} = 1$$

Wednesday 15 April 2026

5.1 Daily Limit

Asymptotic Factorial and Double Factorial Product Limit

Evaluate the limit:

$$\lim_{n\to\infty}\left(\sqrt[n+1]{(n+1)!}-\sqrt[n]{n!}\right)\sqrt[n]{(2n-1)!!}\,\sin\left(\frac{\pi}{\sqrt[n]{n!}}\right)$$

Solution: We analyze each factor in the product separately using Stirling's approximation:

$$\sqrt[n]{n!}=\frac{n}{e}+\frac{\ln(2\pi n)}{2e}+o(1)$$

1. By Lalescu's limit for the difference of consecutive roots of factorials:

$$\lim_{n\to\infty}\left(\sqrt[n+1]{(n+1)!}-\sqrt[n]{n!}\right)=\frac{1}{e}$$

2. For the double factorial factor, we evaluate its asymptotic behavior directly. Using the standard definition $(2n-1)!!=\frac{(2n)!}{2^n n!}$ and applying Stirling's approximation to the factorials:

$$(2n-1)!!\sim\frac{\sqrt{4\pi n}\left(\frac{2n}{e}\right)^{2n}}{2^n\sqrt{2\pi n}\left(\frac{n}{e}\right)^n}=\sqrt{2}\left(\frac{2n}{e}\right)^n$$

Taking the n-th root of this equivalence as $n\to\infty$:

$$\sqrt[n]{(2n-1)!!}\sim\left(\sqrt{2}\right)^{\frac{1}{n}}\frac{2n}{e}\sim\frac{2n}{e}$$

3. For the sine factor, since $\sqrt[n]{n!}\sim\frac{n}{e}\to\infty$, the argument $\frac{\pi}{\sqrt[n]{n!}}\to 0$. Using $\sin(\theta)\sim\theta$:

$$\sin\left(\frac{\pi}{\sqrt[n]{n!}}\right)\sim\frac{\pi}{n/e}=\frac{e\pi}{n}$$

Combining all three asymptotic components:

$$L=\lim_{n\to\infty}\left(\frac{1}{e}\right)\cdot\left(\frac{2n}{e}\right)\cdot\left(\frac{e\pi}{n}\right)=\frac{1}{e}\cdot\frac{2n}{e}\cdot\frac{e\pi}{n}=\frac{2\pi}{e}$$

Thursday 16 April 2026

6.1 Daily Limit

Exponential Difference Quotient Limit

Evaluate the limit:

$$\lim_{x\to 0} \frac{e^x - e^{\sin(x)}}{x - \sin(x)}$$

Solution: Let $u = \sin(x)$. As $x \to 0$, we have $u \to 0$ and $x - \sin(x) \to 0$. We rewrite the numerator by factoring out $e^{\sin(x)}$:

$$e^x - e^{\sin(x)} = e^{\sin(x)} \left(e^{x-\sin(x)} - 1 \right)$$

Substituting this back into the limit quotient:

$$L = \lim_{x\to 0} \frac{e^{\sin(x)} \left(e^{x-\sin(x)} - 1 \right)}{x - \sin(x)}$$

Let $t = x - \sin(x)$. Since $x \neq 0 \implies x \neq \sin(x)$, $t \to 0$ as $x \to 0$. We split the limit into two independent limits:

$$L = \left(\lim_{x\to 0} e^{\sin(x)} \right) \cdot \left(\lim_{t\to 0} \frac{e^t - 1}{t} \right)$$

Evaluating each factor: 1. $\lim_{x\to 0} e^{\sin(x)} = e^{\sin(0)} = e^0 = 1$ by continuity. 2. $\lim_{t\to 0} \frac{e^t-1}{t} = 1$ as the standard exponential derivative limit.

Therefore:

$$L = 1 \cdot 1 = 1$$

Friday 17 April 2026

7.1 Daily Limit

Limit of an Integral of Finite Power Series

Evaluate the limit:

$$\lim_{x\to\infty} \int_{e^{-x}}^{\zeta(x)} \sum_{n=1}^{x} \frac{t^n(1-t^n)}{n}\, dt$$

Solution: As $x \to \infty$, the integration limits behave as follows:

$$e^{-x} \to 0$$

$$\zeta(x) \to 1 \quad \text{since } \lim_{x\to\infty} \zeta(x) = 1$$

For $t \in (0, 1)$, as $x \to \infty$, the finite sum converges to the full power series expansion:

$$\sum_{n=1}^{\infty} \frac{t^n(1-t^n)}{n} = \sum_{n=1}^{\infty} \frac{t^n}{n} - \sum_{n=1}^{\infty} \frac{t^{2n}}{n}$$

Recalling the standard logarithmic power series $\sum_{n=1}^{\infty} \frac{u^n}{n} = -\ln(1-u)$ for $|u| < 1$:

$$\sum_{n=1}^{\infty} \frac{t^n}{n} = -\ln(1-t)$$

$$\sum_{n=1}^{\infty} \frac{t^{2n}}{n} = -\ln(1-t^2)$$

Subtracting the two series gives:

$$f(t) = -\ln(1-t) + \ln(1-t^2) = \ln\left(\frac{1-t^2}{1-t}\right) = \ln(1+t)$$

Now evaluate the definite integral of $\ln(1+t)$ from $t = 0$ to $t = 1$:

$$\int_0^1 \ln(1+t)\, dt = [(1+t)\ln(1+t) - (1+t)]_0^1$$

Evaluating at the boundaries:

$$\text{At } t = 1: \quad 2\ln(2) - 2$$

$$\text{At } t = 0: \quad 1\ln(1) - 1 = -1$$

Subtracting the boundary values:

$$L = (2\ln(2) - 2) - (-1) = 2\ln(2) - 1 = \ln(4) - 1$$

Saturday 18 April 2026

8.1 Daily Limit

Double Riemann Sum Limit

Evaluate the limit:

$$\lim_{n\to\infty} \frac{1}{n^3} \sum_{i=0}^{n} \sum_{j=0}^{n} \sqrt{i^2 + j^2}$$

Solution: We rewrite the double sum as a two-dimensional Riemann sum over the unit square $[0,1] \times [0,1]$:

$$\frac{1}{n^3} \sum_{i=0}^{n} \sum_{j=0}^{n} \sqrt{i^2 + j^2} = \frac{1}{n^2} \sum_{i=0}^{n} \sum_{j=0}^{n} \sqrt{\left(\frac{i}{n}\right)^2 + \left(\frac{j}{n}\right)^2}$$

As $n \to \infty$, this double Riemann sum converges directly to the double integral:

$$I = \int_0^1 \int_0^1 \sqrt{x^2 + y^2}\, dx\, dy$$

Due to the symmetry of the integrand, we convert to polar coordinates (r, θ). The square $[0,1] \times [0,1]$ is split by the diagonal $\theta = \frac{\pi}{4}$ into two symmetric triangles. Evaluating over $0 \le \theta \le \frac{\pi}{4}$:

$$I = 2 \int_0^{\pi/4} \int_0^{\sec\theta} r \cdot r\, dr\, d\theta = 2 \int_0^{\pi/4} \left[\frac{r^3}{3}\right]_0^{\sec\theta} d\theta = \frac{2}{3} \int_0^{\pi/4} \sec^3\theta\, d\theta$$

Using the standard antiderivative formula $\int \sec^3\theta\, d\theta = \frac{1}{2}(\sec\theta\tan\theta + \ln|\sec\theta + \tan\theta|)$:

$$\int_0^{\pi/4} \sec^3\theta\, d\theta = \frac{1}{2}\left[\sqrt{2}(1) + \ln(\sqrt{2}+1)\right] = \frac{\sqrt{2} + \ln(1+\sqrt{2})}{2}$$

Multiplying by $\frac{2}{3}$:

$$I = \frac{2}{3} \cdot \frac{\sqrt{2} + \ln(1+\sqrt{2})}{2} = \frac{\sqrt{2} + \ln(1+\sqrt{2})}{3}$$

Sunday 19 April 2026

9.1 Daily Limit

Limit of Difference of Cube Root and Linear Term

Evaluate the limit:

$$\lim_{n\to\infty}\left(\sqrt[3]{n^3+n^2}-n\right)$$

Solution: We factor out n^3 from inside the cube root:

$$\sqrt[3]{n^3+n^2}=\sqrt[3]{n^3\left(1+\frac{1}{n}\right)}=n\left(1+\frac{1}{n}\right)^{\frac{1}{3}}$$

Now, apply the generalized binomial expansion $(1+u)^\alpha=1+\alpha u+\frac{\alpha(\alpha-1)}{2}u^2+O(u^3)$ with $\alpha=\frac{1}{3}$ and $u=\frac{1}{n}\to 0$:

$$\left(1+\frac{1}{n}\right)^{\frac{1}{3}}=1+\frac{1}{3n}-\frac{1}{9n^2}+O\left(\frac{1}{n^3}\right)$$

Multiplying through by n:

$$n\left(1+\frac{1}{n}\right)^{\frac{1}{3}}=n+\frac{1}{3}-\frac{1}{9n}+O\left(\frac{1}{n^2}\right)$$

Subtracting n from both sides:

$$\sqrt[3]{n^3+n^2}-n=\frac{1}{3}-\frac{1}{9n}+O\left(\frac{1}{n^2}\right)$$

Taking the limit as $n\to\infty$:

$$L=\lim_{n\to\infty}\left(\frac{1}{3}-\frac{1}{9n}+O\left(\frac{1}{n^2}\right)\right)=\frac{1}{3}$$

Monday 20 April 2026

10.1 Daily Limit

Nested Inverse Trigonometric and Hyperbolic Functions Limit

Evaluate the limit:

$$\lim_{x \to 0} \frac{\arcsin(\arctan(\operatorname{arcsinh}(\operatorname{arctanh}(x))))}{\ln(\Gamma(1+x))}$$

Solution: We evaluate the asymptotic expansions of the numerator and denominator around $x = 0$ as $x \to 0$.

1. **Numerator:** Recall the Maclaurin series for each inverse function near 0:

$$\operatorname{arctanh}(x) = x + O(x^3)$$

$$\operatorname{arcsinh}(u) = u + O(u^3) \implies \operatorname{arcsinh}(\operatorname{arctanh}(x)) = x + O(x^3)$$

$$\arctan(v) = v + O(v^3) \implies \arctan(\operatorname{arcsinh}(\operatorname{arctanh}(x))) = x + O(x^3)$$

$$\arcsin(w) = w + O(w^3) \implies \arcsin(\arctan(\operatorname{arcsinh}(\operatorname{arctanh}(x)))) = x + O(x^3)$$

Thus, the numerator is simply $x + O(x^3)$.

2. **Denominator:** Recall the Taylor expansion for $\ln(\Gamma(1+x))$ near $x = 0$:

$$\ln(\Gamma(1+x)) = -\gamma x + \frac{\pi^2}{12} x^2 + O(x^3)$$

where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant (since $\psi(1) = \Gamma'(1) = -\gamma$).

Combining the numerator and denominator expansions:

$$L = \lim_{x \to 0} \frac{x + O(x^3)}{-\gamma x + O(x^2)} = \lim_{x \to 0} \frac{1 + O(x^2)}{-\gamma + O(x)} = -\frac{1}{\gamma}$$